

OCR B A-Level Chemistry — Practical Activities

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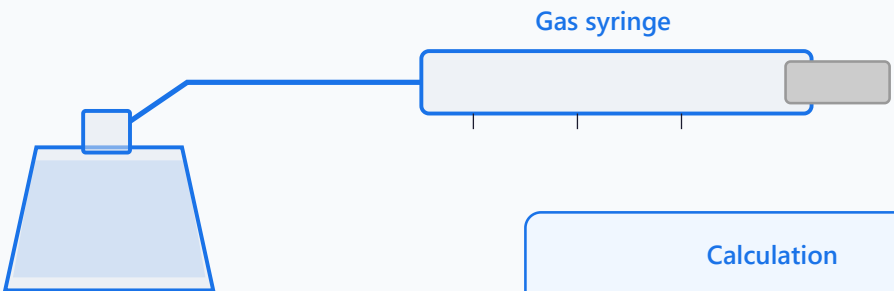
Supporting practicals (not PAGs):

- [Colorimetry and Transition Metal Complexes](#)
- [Chromatography](#)
- [Equilibrium Investigation](#)

OCR Practical Endorsement: the Chemistry B (Salters) specification (H433, section 5h) uses the same twelve Practical Activity Groups as Chemistry A, PAG1–PAG12. You must complete a minimum of 12

practical activities covering all of them. The experiment in each section below is one suitable example; your school may use a different activity for the same PAG.

PAG1: Moles Determination



Gas syringe

Calculation

$$n(\text{Mg}) = \text{mass} / A_r$$
$$n(\text{H}_2) = n(\text{Mg}) \text{ [1:1 ratio]}$$
$$V(\text{gas}) = \text{reading from syringe}$$
$$\text{Molar volume} = V / n$$

Balance: 0.024 g Mg

Expected: $\sim 24,000 \text{ cm}^3 \text{ mol}^{-1}$ at RTP (room temperature and pressure)

Apparatus

- Conical flask with bung and delivery tube
- Gas syringe (100 cm^3)
- Balance (to 0.001 g)
- Measuring cylinder
- Magnesium ribbon (weighed accurately)
- Excess dilute HCl
- Thermometer and barometer (to record conditions)

Method

1. Weigh a known length of clean Mg ribbon accurately (e.g., ~0.03 g).
2. Place excess dilute HCl (e.g., 20 cm³ of 1 mol dm⁻³) in the conical flask.
3. Connect the delivery tube to the gas syringe. Check there are no leaks.
4. Add the Mg ribbon to the flask, quickly insert the bung, and allow the reaction to proceed.
5. Wait until the reaction is complete (Mg fully dissolved, no more gas produced).
6. Record the final volume of H₂ gas in the syringe.
7. Record room temperature and atmospheric pressure.
8. Calculate: $n(\text{Mg}) = \text{mass}/24.3$. $n(\text{H}_2) = n(\text{Mg})$. Molar gas volume = $V(\text{gas})/n(\text{H}_2)$.
9. Alternatively, use $pV = nRT$ to calculate n from p , V , and T , then find M_r .

Safety

- H₂ gas is flammable — no naked flames.
- Dilute HCl is irritant — goggles.
- Gas syringe may pop out — clamp securely.

Analysis

Calculate: Molar gas volume at room conditions. Expected ~24 dm³ mol⁻¹ (24,000 cm³ mol⁻¹) at ~298 K, 101 kPa.

Alternative: Use $pV = nRT$ to find n , then compare with calculated n from mass.

Error Analysis

Source of Error	Type	Improvement
H ₂ gas escaping before sealing	Systematic	Seal quickly, or hang the Mg in a small tube inside the sealed flask and tip it into the acid
Gas syringe friction	Systematic	Lubricate; tap gently
Mg oxide layer on ribbon	Systematic	Clean with emery paper before weighing

Gas not at room temperature	Systematic	Allow time for gas to cool before reading
H ₂ slightly soluble in water/acid	Systematic	Negligible at normal conditions; acknowledged

Exam Tip: The molar gas volume at RTP (room temperature and pressure, ~298 K, 100 kPa) is approximately 24.0 dm³ mol⁻¹. At STP (273 K, 100 kPa) it is 22.7 dm³ mol⁻¹. Always state the conditions. Use $pV = nRT$ when exact conditions matter: p in Pa, V in m³, T in K, $R = 8.314$.

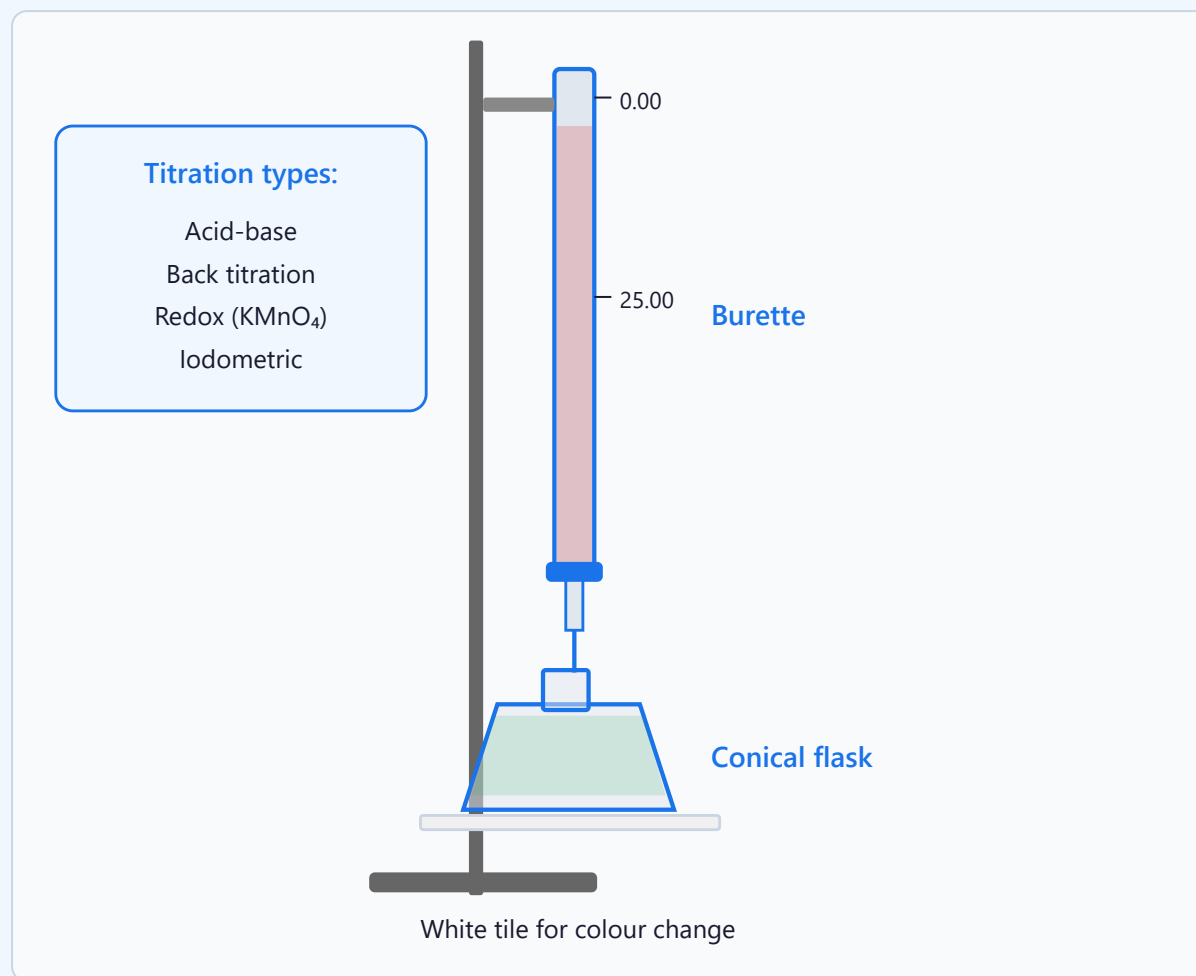
Worked Example — Molar Gas Volume:

0.0360 g of magnesium ribbon reacted with excess dilute HCl, producing 35.8 cm³ of H₂ at room temperature and pressure.

$$n(\text{Mg}) = 0.0360 / 24.3 = 1.48 \times 10^{-3} \text{ mol} = n(\text{H}_2) \text{ (1:1 ratio)}$$

Molar gas volume = $35.8 / 1.48 \times 10^{-3} = 2.42 \times 10^4 \text{ cm}^3 \text{ mol}^{-1} = \mathbf{24.2 \text{ dm}^3 \text{ mol}^{-1}}$, close to the expected 24.0 dm³ mol⁻¹ at RTP.

PAG2: Acid–Base Titration



Apparatus

- 50 cm^3 burette, clamp and stand
- 25.0 cm^3 pipette and filler
- Conical flasks (250 cm^3), white tile
- Suitable indicator or self-indicating reagent
- Volumetric flask (for preparing standard solutions)

Method

1. **Standard solution:** Weigh the solid accurately (weigh the boat before and after transfer), dissolve in distilled water in a beaker, transfer with rinsings to a 250 cm^3 volumetric flask, make up to the mark (bottom of meniscus at eye level) and invert to mix.
2. **Acid-base:** Pipette 25.0 cm^3 of one solution into a conical flask. Add indicator. Titrate from the burette until permanent colour change. Repeat for concordant results (within 0.10 cm^3).

3. **Back titration:** React the sample with known excess of reagent. Titrate the remaining unreacted reagent. Calculate original amount by difference. Useful when the analyte is insoluble (e.g., CaCO_3 in antacid tablets).
4. **Indicator choice:** strong acid/strong base — methyl orange or phenolphthalein; strong acid/weak base (e.g. HCl with NaHCO_3) — methyl orange; weak acid/strong base — phenolphthalein.
5. **Extension — Redox (KMnO_4):** redox titrations use the same technique and are examinable, but PAG2 itself is an acid–base titration. Pipette acidified Fe^{2+} into flask. Titrate with KMnO_4 until permanent faint pink (self-indicating). Use H_2SO_4 to acidify (not HCl or HNO_3).
6. Calculate using $n = cV$ and stoichiometric ratios from balanced equations.

Safety

- Acids and alkalis — corrosive/irritant, goggles.
- KMnO_4 — oxidising agent, stains; wear gloves.
- Use pipette filler — never mouth-pipette.

Analysis

Calculate: Mean concordant titre. $n(\text{known}) = c \times V$. Use stoichiometry for $n(\text{unknown})$. $c = n/V$.

Back titration: $n(\text{original}) = n(\text{added}) - n(\text{remaining after back-titration})$.

Error Analysis

Source of Error	Type	Improvement
Standard solution not made up exactly to the mark	Systematic	Rinse all solid into the flask; add the last drops with a dropping pipette
Overshooting end point	Systematic	Add drop-wise near end point
Parallax reading burette	Random	Eye level; bottom of meniscus (top for KMnO_4)
Not rinsing burette correctly	Systematic	Rinse with solution to be used

Including rough titre in mean

Systematic

Only average concordant results; exclude rough

Exam Tip: For back titrations, always set up the calculation as: moles reacted with analyte = total moles added – moles determined by titration. Common example: finding % CaCO_3 in limestone — add excess HCl, then titrate remaining HCl with NaOH.

Worked Example — Back Titration:

1.50 g of impure CaCO_3 was dissolved in 50.0 cm^3 of 1.00 mol dm^{-3} HCl (excess). The remaining HCl required 23.05 cm^3 of 1.00 mol dm^{-3} NaOH to neutralise it.

$$n(\text{HCl total}) = 1.00 \times 0.0500 = 0.0500 \text{ mol}$$

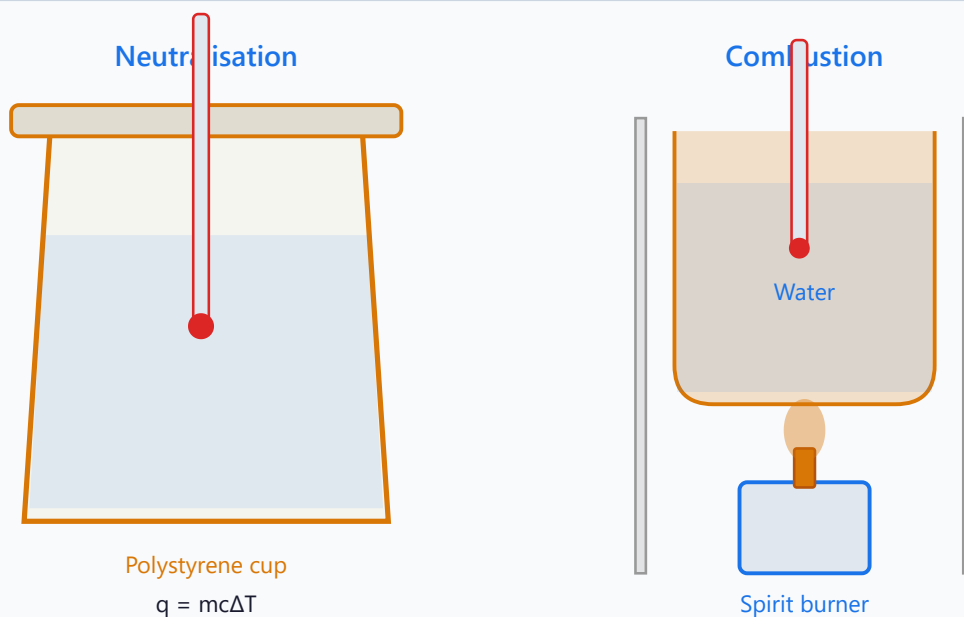
$$n(\text{NaOH}) = 1.00 \times 0.02305 = 0.02305 \text{ mol} = n(\text{HCl remaining})$$

$$n(\text{HCl reacted with CaCO}_3) = 0.0500 - 0.02305 = 0.02695 \text{ mol}$$

$$n(\text{CaCO}_3) = 0.02695 / 2 = 0.01348 \text{ mol} \quad (\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2)$$

$$\text{Mass CaCO}_3 = 0.01348 \times 100.1 = 1.349 \text{ g} \rightarrow \text{percentage purity} = (1.349 / 1.50) \times 100 = \mathbf{89.9\%}$$

PAG3: Enthalpy Determination



Neutralisation: $\Delta H \approx -57 \text{ kJ mol}^{-1}$ (strong acid + strong base)

Combustion: $\Delta H = -q/n$ (always less exothermic than literature due to heat loss)

Apparatus

- **Neutralisation:** Polystyrene cup, lid, thermometer, measuring cylinder, stopwatch
- **Combustion:** Spirit burner, copper calorimeter, thermometer, balance, draught shield, measuring cylinder

Method (Neutralisation)

1. Measure 25 cm^3 of acid into the polystyrene cup. Record temperature for 3 minutes.
2. At minute 4, add 25 cm^3 of alkali. Stir. Record temperature for 6+ minutes.
3. Use graphical extrapolation to find ΔT . Calculate $q = mc\Delta T$. Then $\Delta H = -q/n$.

Method (Combustion)

1. Measure 100 cm^3 of water into the calorimeter. Record initial temperature.
2. Weigh the spirit burner. Light it and heat the water until temperature rises $\sim 20^\circ \text{C}$.
3. Extinguish, reweigh. Calculate mass burned, $q = mc\Delta T$, $n = \text{mass}/M_r$. $\Delta H = -q/n$.

Safety

- Acids/alkalis — corrosive, goggles.
- Alcohols are flammable — keep away from other ignition sources.
- Hot apparatus — use tongs.

Analysis

Plot: Temperature vs. time for extrapolation (neutralisation).

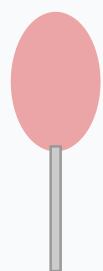
Compare: Experimental vs. literature ΔH values. Discuss reasons for discrepancy.

Error Analysis

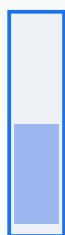
Source of Error	Type	Improvement
Heat loss to surroundings	Systematic	Lid, insulation, draught shield, extrapolation
Incomplete combustion	Systematic	Ensure good air supply; clean wick
Evaporation of alcohol	Systematic	Cap promptly; weigh quickly
Heat capacity assumptions	Systematic	Acknowledged; dilute solution $\approx$ water

Exam Tip: The experimental $\Delta H(\text{combustion})$ is always less negative (less exothermic) than the data book value. The main reason is heat loss to surroundings. Other factors include incomplete combustion and heat absorbed by the apparatus. Always state these clearly for full marks.

PAG4: Qualitative Analysis of Ions



Flame tests
Li Na K Ca Ba



Cu^{2+}



Fe^{2+}



Fe^{3+}

NaOH precipitates

Anion Tests

Cl^- : $\text{AgNO}_3 \rightarrow$ white ppt

Br^- : $\text{AgNO}_3 \rightarrow$ cream ppt

I^- : $\text{AgNO}_3 \rightarrow$ yellow ppt

SO_4^{2-} : $\text{BaCl}_2 \rightarrow$ white ppt

CO_3^{2-} : $\text{HCl} \rightarrow \text{CO}_2$ (limewater)

NH_4^+ : $\text{NaOH} + \text{heat} \rightarrow \text{NH}_3$

Always acidify with HNO_3 before AgNO_3 and BaCl_2 tests

Apparatus

- Test tubes, nichrome wire, Bunsen burner
- $\text{NaOH}(\text{aq})$, $\text{NH}_3(\text{aq})$, $\text{AgNO}_3(\text{aq})$, $\text{BaCl}_2(\text{aq})$ or $\text{Ba}(\text{NO}_3)_2(\text{aq})$, dilute HNO_3 , HCl , Devarda's alloy
- Limewater, litmus paper

Method

1. **Flame tests:** Clean wire $\rightarrow$ dip in sample $\rightarrow$ observe colour in Bunsen flame.
2. **Cation tests:** Add $\text{NaOH}(\text{aq})$ dropwise, then in excess — observe precipitate colour and whether it redissolves (e.g. Cu^{2+} blue, Fe^{2+} green, Fe^{3+} orange-brown; Zn^{2+} , Al^{3+} and Pb^{2+} white precipitates that dissolve in excess NaOH).
3. **Halide tests:** Acidify with HNO_3 , add AgNO_3 . Confirm with NH_3 : AgCl dissolves in dilute NH_3 , AgBr in concentrated NH_3 , AgI does not dissolve.
4. **Sulfate:** Acidify with HNO_3 , add $\text{BaCl}_2 \rightarrow$ white precipitate.
5. **Carbonate:** Add dilute HNO_3 (not HCl if you will test the same sample for halides) $\rightarrow$ effervescence, gas turns limewater milky.
6. **Ammonium:** Add NaOH , warm $\rightarrow \text{NH}_3$ turns damp red litmus blue.
7. **Nitrate(V):** Warm with $\text{NaOH}(\text{aq})$ and Devarda's alloy $\rightarrow \text{NH}_3$ given off (turns damp red litmus blue).

8. **Sensible order on one sample:** carbonate, then sulfate, then halide — BaCO_3 is insoluble and Ag_2SO_4 is only slightly soluble, so either could give a false positive; use $\text{Ba}(\text{NO}_3)_2$ rather than BaCl_2 if the halide test follows.

Safety

- AgNO_3 stains — gloves. BaCl_2 toxic — wash hands.
- Conc. HCl and NH_3 — fume cupboard.

Analysis

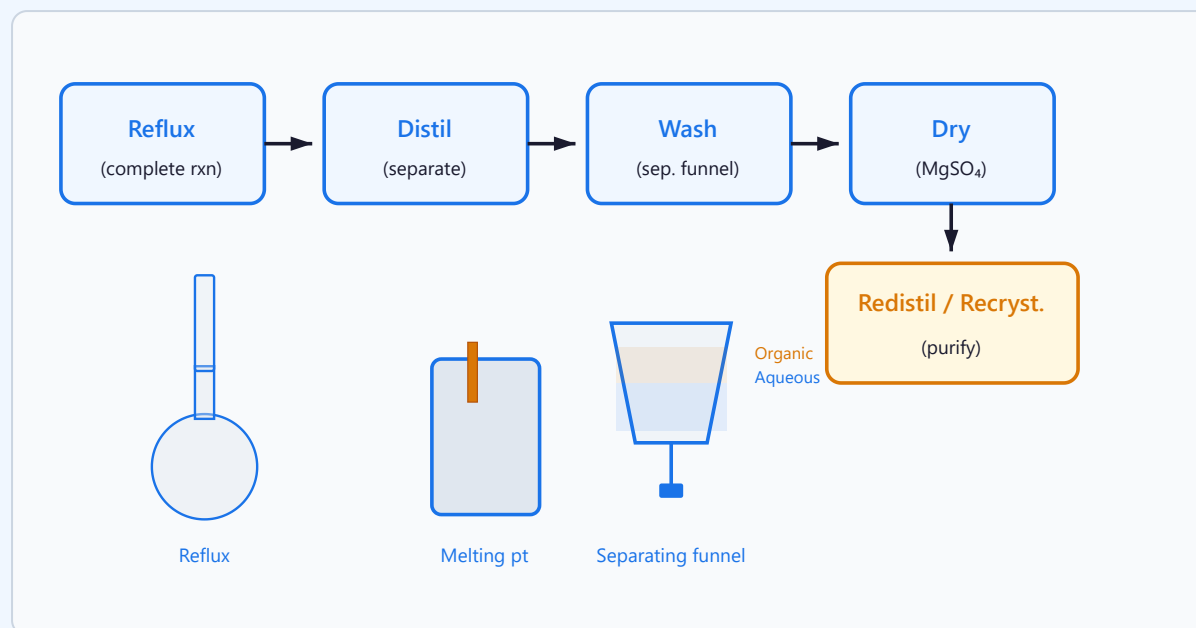
Record: Observations table for each test. Match results to identify unknown ions.

Error Analysis

Source of Error	Type	Improvement
Contamination	Systematic	Clean all equipment; use separate droppers
Na^+ masking flame colours	Systematic	View through cobalt blue glass
Not acidifying before tests	Systematic	Always acidify with HNO_3
Colour confusion (cream vs white)	Random	Compare alongside known standards

Exam Tip: To distinguish between halide silver precipitates further: add dilute NH_3 — AgCl dissolves (soluble complex forms); add conc. NH_3 — AgBr dissolves. AgI does not dissolve in either. This is a common extension question.

PAG5: Synthesis of an Organic Liquid



Apparatus

- Round-bottomed flask, anti-bumping granules
- Reflux condenser (vertical) and distillation condenser (angled)
- Separating funnel, conical flask
- Drying agent (anhydrous MgSO_4 , Na_2SO_4 or CaCl_2)
- Thermometer for the still head; electric heating mantle
- Example reagents (haloalkane): butan-1-ol, sodium bromide, concentrated sulfuric acid, sodium hydrogencarbonate solution

Method (e.g. 1-bromobutane from butan-1-ol)

1. **Reaction:** Heat butan-1-ol with sodium bromide and concentrated sulfuric acid (which generate HBr) under reflux, with anti-bumping granules.
2. **Separation:** Distil to collect the crude product. Wash in a separating funnel with water, then sodium hydrogencarbonate solution (removes acid; vent often), then water. 1-bromobutane is denser than water, so keep the *lower* layer.
3. **Drying:** Add anhydrous drying agent until the liquid is clear. Decant or filter.
4. **Purification:** Redistil, collecting the fraction that boils close to the data-book value (about 100–104 °C).

5. Calculate percentage yield.

Safety

- Organic chemicals are flammable — electric heating, no flames.
- Work in fume cupboard.
- Vent separating funnel to release pressure.
- Concentrated sulfuric acid is corrosive — add slowly with cooling. A written risk assessment is part of this PAG.

Analysis

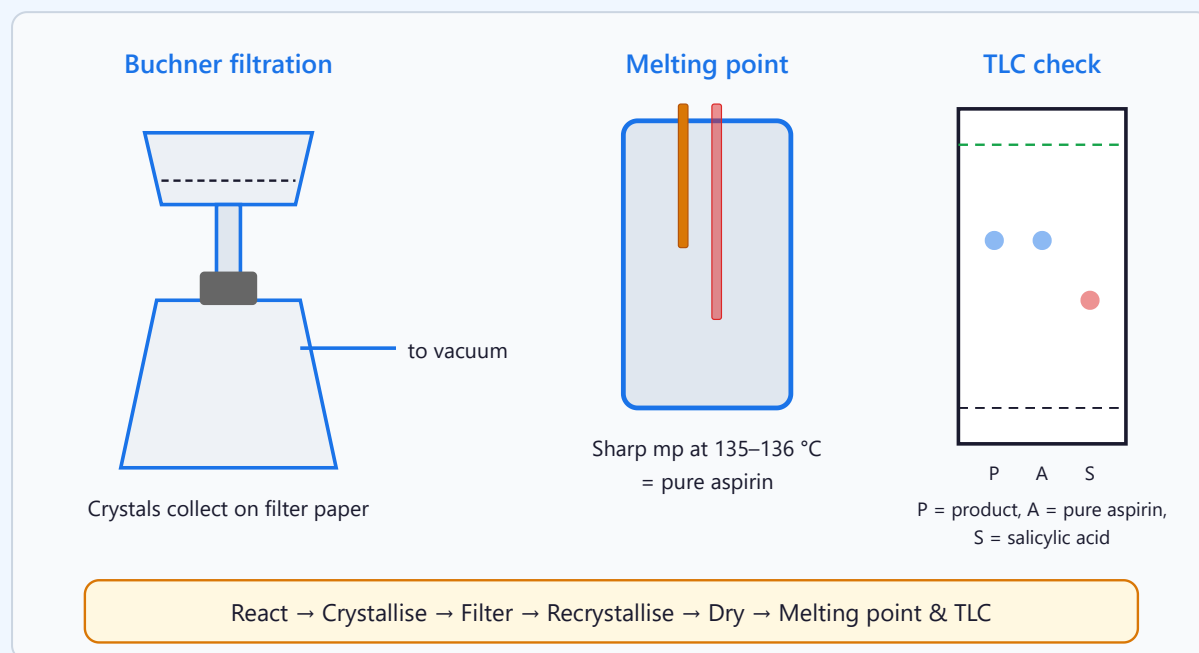
Calculate: % yield = (actual/theoretical) × 100. % atom economy = (M_r desired product / total M_r all products) × 100.

Error Analysis

Source of Error	Type	Improvement
Product lost in transfers	Systematic	Minimise transfers; rinse with small amounts
Incomplete reaction	Systematic	Reflux longer; use excess reagent
Product still wet (cloudy)	Systematic	Add more drying agent until the liquid is clear
Wrong fraction collected	Systematic	Monitor thermometer; collect narrow range

Exam Tip: Always explain the purpose of each step: reflux = complete reaction; distillation = separate by boiling point; washing with NaHCO_3 (or Na_2CO_3) = removes acidic impurities; drying agent = removes water; redistillation = purifies the product, collected at its boiling point. Never heat a sealed system. Each step reduces yield, so high yields require careful technique.

PAG6: Synthesis of an Organic Solid



Apparatus

- Conical flask or pear-shaped flask (with reflux condenser if heating for longer), water bath or electric heater
- Salicylic acid (2-hydroxybenzoic acid), ethanoic anhydride, a few drops of concentrated phosphoric(V) or sulfuric acid as catalyst
- Buchner funnel, filter flask, filter paper, vacuum pump; ice bath
- Solvent for recrystallisation (e.g. hot water or an ethanol–water mixture)
- Melting point apparatus and capillary tubes
- TLC plate, developing chamber, capillary spotters, UV lamp

Method (Synthesis of Aspirin)

1. Weigh about 2 g of salicylic acid accurately into a dry flask. Add excess ethanoic anhydride (about 4 cm³) in a fume cupboard, then a few drops of the acid catalyst.
2. Warm in a water bath at roughly 80–90 °C for 10–15 minutes, swirling, until all the solid has dissolved. (Heating under reflux can be used instead; reflux may be covered in either PAG5 or

PAG6.)

- Carefully add cold water to hydrolyse the excess ethanoic anhydride, then cool the flask in an ice bath to crystallise the crude aspirin.
- Collect the crystals by suction filtration using a Buchner funnel. Wash with a little ice-cold water.
- Recrystallise:** dissolve the crude solid in the minimum volume of hot solvent, filter while hot if there are insoluble impurities, then cool slowly so that pure crystals form and soluble impurities stay in solution. Filter again under reduced pressure, wash with a little cold solvent, and dry (e.g. between filter papers or in a desiccator).
- Weigh the dry product and calculate the percentage yield.
- Check purity:** measure the melting point range and compare with the data-book value (about 136 °C). Run a TLC plate spotted with the product, pure aspirin and salicylic acid.

Safety

- Ethanoic anhydride is corrosive and its vapour is irritating to the eyes and lungs — dispense in a fume cupboard.
- Concentrated phosphoric(V)/sulfuric acid is corrosive — goggles and gloves.
- Ethanol (if used in recrystallisation) is flammable — heat with a water bath or electric heater, not a flame.
- A written risk assessment is part of this PAG: identify each hazard, the risk and how it is controlled.

Analysis

Calculate: Percentage yield = (actual mass / theoretical mass) × 100, using the limiting reagent (salicylic acid; ethanoic anhydride is in excess).

Purity: A pure solid melts sharply (over 1–2 °C) at the data-book value. On the TLC plate, a pure product gives a single spot at the same R_f as pure aspirin; a spot matching salicylic acid shows unreacted starting material.

Error Analysis

Source of Error	Type	Improvement
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Too much solvent used in recrystallisation	Systematic	Use the minimum volume of hot solvent so less product stays dissolved
Product lost during transfers and filtration	Systematic	Minimise transfers; rinse with small amounts of ice-cold solvent
Wet product (mass too high; mp lowered)	Systematic	Dry thoroughly; dry to constant mass
Heating melting point apparatus too quickly	Systematic	Heat slowly (about 1–2 °C per minute) near the expected mp

Exam Tip: Be ready to explain each recrystallisation step: *minimum* hot solvent (so the product crystallises on cooling), hot filtration (removes insoluble impurities), slow cooling (purer, larger crystals), cold-solvent wash (removes soluble impurities on the crystal surface without redissolving much product). Impurities *lower and broaden* the melting point range.

Worked Example — Percentage Yield:

A student prepared aspirin from 2.00 g of salicylic acid. The theoretical yield is 2.61 g. The actual yield after purification was 1.85 g.

$$\text{Percentage yield} = (1.85 / 2.61) \times 100 = \mathbf{70.9\%}$$

This is a typical yield for a lab preparation. Losses occur during transfer, filtration, washing, and recrystallisation.

PAG7: Qualitative Analysis of Organic Functional Groups

Alkene (C=C) $\text{Br}_2(\text{aq})$ decolourises

Orange → colourless

Aldehyde (CHO)

Tollens' → Ag mirror

Fehling's → red ppt

Amine ($-\text{NH}_2$)

Indicator paper

Alkaline solution

Alcohol (OH) $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$ warmOrange → green ($1^\circ/2^\circ$)**Carboxylic acid** $\text{Na}_2\text{CO}_3 \rightarrow$ fizzes (CO_2)

Limewater goes milky

PhenolNeutral $\text{FeCl}_3(\text{aq})$

Purple colour

Warm Tollens'/Fehling's/ $\text{K}_2\text{Cr}_2\text{O}_7$ in a water bath at 60-70 °C

Apparatus

- Test tubes, water bath (60-70 °C)
- Bromine water, Tollens' reagent, Fehling's A+B
- Acidified $\text{K}_2\text{Cr}_2\text{O}_7$, Na_2CO_3 solution, limewater, neutral iron(III) chloride solution, indicator paper

Method

1. **C=C:** Add $\text{Br}_2(\text{aq})$ — decolourises = alkene.
2. **CHO (aldehyde):** Tollens' (silver mirror) or Fehling's (brick-red precipitate).
3. **OH (alcohol $1^\circ/2^\circ$):** Warm with acidified $\text{K}_2\text{Cr}_2\text{O}_7$ — orange to green.
4. **COOH:** Add Na_2CO_3 — effervescence = carboxylic acid.
5. **Phenol:** Neutral $\text{FeCl}_3(\text{aq})$ — purple colour. Phenols are weakly acidic (dissolve in NaOH) but do *not* fizz with Na_2CO_3 .
6. **Amine:** Solution turns damp indicator paper alkaline (lone pair on N accepts H^+).

Safety

- $\text{K}_2\text{Cr}_2\text{O}_7$ is carcinogenic — gloves, fume cupboard.
- Tollens' — prepare fresh, dispose immediately (explosive on standing).
- Bromine water is toxic and corrosive — fume cupboard.

Analysis

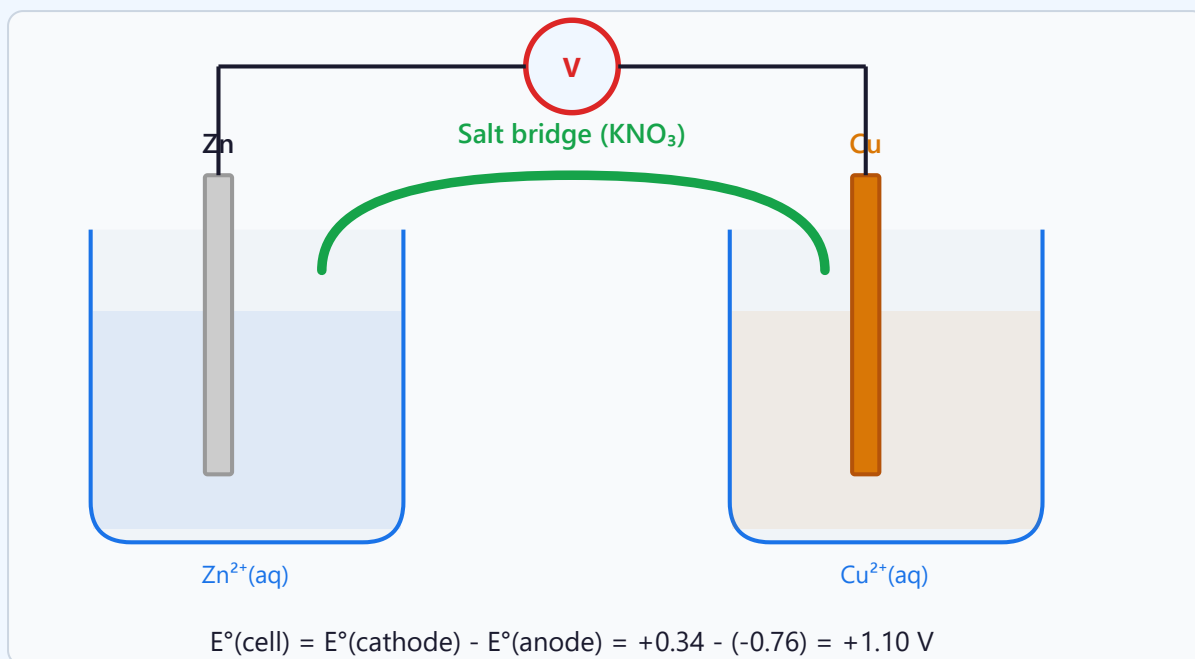
Record: Table of positive/negative results. Use the pattern to identify functional groups.

Error Analysis

Source of Error	Type	Improvement
Dirty glassware	Systematic	Clean thoroughly; Tollens' needs very clean tubes
Insufficient heating	Systematic	Use water bath at correct temperature
Using old Tollens' reagent	Systematic	Prepare fresh immediately before use
False positives from contamination	Systematic	Run blank tests; use fresh reagents

Exam Tip: To distinguish aldehyde from ketone: only aldehydes give a silver mirror with Tollens' or a red precipitate with Fehling's; ketones give no change. To distinguish a phenol from a carboxylic acid: both are acidic, but only the carboxylic acid fizzes with carbonate, and only the phenol gives a purple colour with neutral FeCl_3 . Tertiary alcohols do NOT react with $\text{K}_2\text{Cr}_2\text{O}_7$ (no colour change) because they cannot be oxidised without breaking C-C bonds.

PAG8: Electrochemical Cells



Apparatus

- Beakers, metal electrodes (Zn, Cu, Fe, Mg, etc.)
- 1.0 mol dm^{-3} metal salt solutions
- Salt bridge (KNO_3 soaked filter paper or agar gel)
- High-resistance voltmeter, wires, crocodile clips
- Emery paper

Method

1. Clean electrodes with emery paper to remove oxide.
2. Place each electrode in its corresponding 1.0 mol dm^{-3} metal ion solution.
3. Connect the two half-cells with a salt bridge.
4. Connect electrodes to the high-resistance voltmeter.
5. Record the EMF and note the polarity (which is +, which is -).
6. Test multiple metal combinations and compare with E° values.
7. Investigate the effect of concentration: dilute one electrolyte (e.g. the Cu^{2+} solution in a Zn/Cu cell) and record the new cell potential.

Safety

- Metal salt solutions may be toxic — gloves and goggles.
- Dispose of solutions correctly.

Analysis

Calculate: $E^\circ(\text{cell}) = E^\circ(\text{cathode}) - E^\circ(\text{anode})$. Compare measured with theoretical.

Investigate: Effect of changing concentration on EMF (qualitative Nernst).

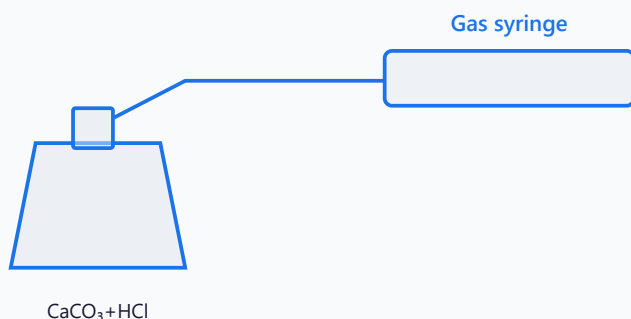
Error Analysis

Source of Error	Type	Improvement
Oxide layer on electrodes	Systematic	Clean with emery paper before use
Non-standard concentrations	Systematic	Use volumetric flask to prepare
Salt bridge drying out	Systematic	Fresh salt bridge; good immersion
Voltmeter not high enough resistance	Systematic	Use high-resistance voltmeter or potentiometer

Exam Tip: The more reactive metal is the anode (negative terminal, oxidation). The less reactive is the cathode (positive terminal, reduction). A high-resistance voltmeter is essential because if current flows, the cell does work and the measured EMF is less than the true EMF. Remember: AN OX (anode = oxidation), RED CAT (reduction = cathode).

PAG9: Rates of Reaction — Continuous Monitoring Method

Continuous Monitoring



Plot volume vs time | tangent gradient = rate | constant half-life = 1st order

Apparatus

- **Continuous:** Conical flask, gas syringe, delivery tube, stopwatch, measuring cylinder
- **Alternatives:** top-pan balance (mass of CO_2 lost) or colorimeter with data logger (colour change)

Method

1. **Continuous monitoring:** Collect gas at timed intervals from a reaction (e.g., $\text{CaCO}_3 + \text{HCl}$). Plot volume vs. time. Initial rate = gradient at $t = 0$.
2. Take readings at regular intervals (e.g. every 30 s) until the reaction stops, keeping the temperature constant.
3. Convert readings into the amount of reactant remaining (proportional to $V_{\text{final}} - V$) and plot concentration against time to find half-lives.

Safety

- HCl is irritant — goggles.
- Acid spray — use a cotton wool plug if measuring mass loss.

Analysis

Continuous: Tangent at $t = 0$ gives initial rate. Compare different conditions.

Half-life: Constant successive half-lives = first order; then $k = \ln 2 / t_{1/2}$. A straight-line concentration–time graph = zero order.

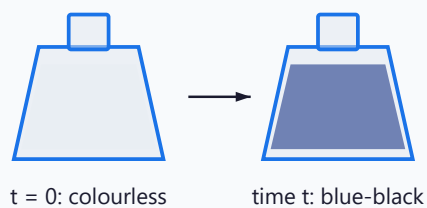
Error Analysis

Source of Error	Type	Improvement
Gas lost before sealing	Systematic	Seal quickly
Reading syringe and clock at the same moment	Random	Two people, or use a data logger
Temperature variation	Systematic	Thermostatted water bath
Volume measurement error	Random	Use burettes for better precision

Exam Tip: Continuous monitoring follows one reaction mixture all the way through, so a single run gives a whole concentration–time curve. Draw tangents to find the rate at any time. For a first-order reaction the half-life is constant and does not depend on the starting concentration, so measuring two or three successive half-lives is a quick check of order.

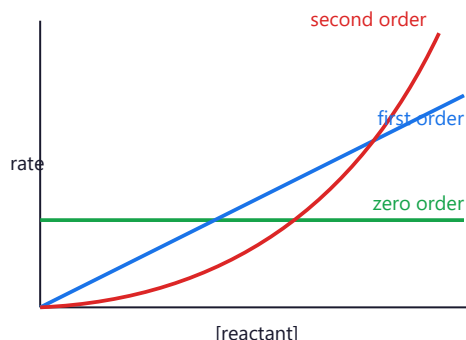
PAG10: Rates of Reaction — Initial Rates Method

Iodine clock



Initial rate $\propto 1/t$

Initial rate vs concentration



Change ONE concentration at a time; keep total volume and temperature constant

$$\text{rate} = k[A]^n[B]^m \rightarrow k = \text{rate} / ([A]^n[B]^m)$$

Apparatus

- Conical flasks or boiling tubes, burettes or graduated pipettes for each solution
- Stopwatch; thermometer; water baths (for the temperature extension)
- **Iodine clock:** potassium iodide, dilute sulfuric acid, hydrogen peroxide, sodium thiosulfate (small fixed amount), starch solution, distilled water
- **Temperature extension:** sodium thiosulfate and dilute hydrochloric acid, paper marked with a cross
- Spreadsheet software to process the results

Method

1. Mix fixed volumes of KI(aq) , $\text{H}_2\text{SO}_4\text{(aq)}$, $\text{Na}_2\text{S}_2\text{O}_3\text{(aq)}$ and starch in a flask. Add the $\text{H}_2\text{O}_2\text{(aq)}$ last and start the stopwatch immediately.
2. The iodine formed ($\text{H}_2\text{O}_2 + 2\text{I}^- + 2\text{H}^+ \rightarrow \text{I}_2 + 2\text{H}_2\text{O}$) is immediately turned back into I^- by the thiosulfate. When all the thiosulfate has been used, free I_2 appears and the starch suddenly turns blue-black. Record this time, t .
3. Repeat with different volumes of KI(aq) , making up the difference with distilled water so that the **total volume stays the same** and only $[\text{I}^-]$ changes.

4. Repeat the series varying $[H_2O_2]$ and then $[H^+]$ in the same way, each time keeping the other concentrations fixed.
5. Carry out each run at the same temperature (measure it) and repeat runs to check reliability.
6. **Temperature extension:** time the disappearance of a cross under a flask of $Na_2S_2O_3(aq) + HCl(aq)$ at several temperatures, keeping all volumes and concentrations the same.

Safety

- Dilute sulfuric acid and hydrochloric acid are irritants — goggles.
- Hydrogen peroxide is an irritant/oxidiser — avoid skin contact.
- The thiosulfate/acid reaction releases SO_2 — work in a well-ventilated room and dispose of the mixture promptly; take care if you have asthma.
- Hot water baths — care with scalds.

Analysis

Orders: Because the same small amount of I_2 is formed before the colour appears, the initial rate $\propto 1/t$. Plot $1/t$ against the concentration you varied: a horizontal line = zero order, a straight line through the origin = first order, an upward curve = second order (confirm by plotting $1/t$ against concentration²). Alternatively plot $\log(\text{rate})$ against $\log(\text{concentration})$: the gradient is the order.

Rate constant: Once the orders are known, substitute an initial rate and the concentrations into the rate equation to find k , and work out its units.

Temperature: Plot $\ln(1/t)$ ($\propto \ln k$) against $1/T$ (T in K). The gradient = $-E_a/R$, so $E_a = -\text{gradient} \times 8.314$.

Error Analysis

Source of Error	Type	Improvement
Judging the exact moment of colour change	Random	Same observer; white background; repeat and average; use a colorimeter/light sensor
Delay between mixing and starting the clock	Random	Add the last reagent quickly and start timing at the same moment

Total volume not kept constant	Systematic	Make up each run to the same total volume with distilled water
Temperature varies between runs	Systematic	Use a thermostatted water bath; pre-equilibrate the solutions

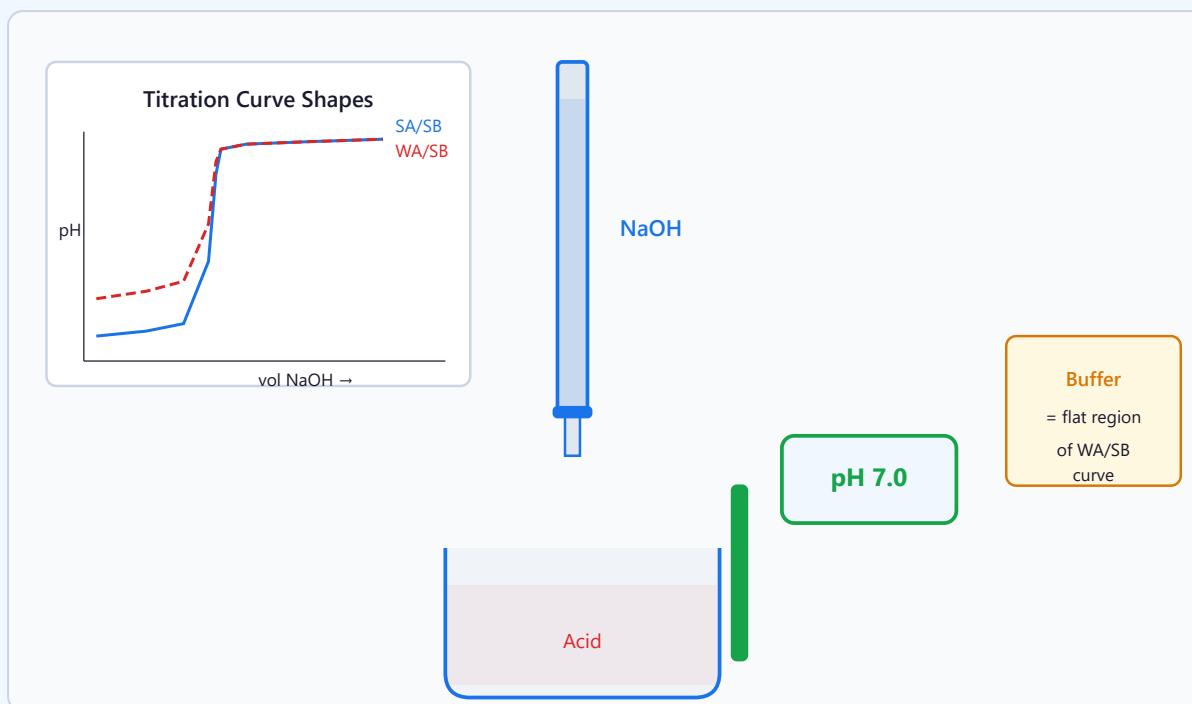
Exam Tip: $1/t$ is only a fair measure of the *initial* rate if the reaction has gone only a small way when the colour appears — that is why only a small, fixed amount of thiosulfate is used. Always state that you keep the concentrations of the other reactants, the total volume and the temperature constant. The units of k depend on the overall order: for rate = $k[A][B]$, k has units $\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$.

Worked Example — Finding an Order:

Doubling $[\text{I}^-]$ (all else constant) changes the time from 84 s to 42 s, so $1/t$ (the initial rate) doubles → the reaction is **first order with respect to I^-** .

Doubling $[\text{H}^+]$ leaves the time unchanged at 84 s → **zero order with respect to H^+** (H^+ does not appear in the rate equation).

PAG11: pH Measurement



Apparatus

- Calibrated pH meter/probe
- Burette with NaOH (or HCl)
- Beaker with acid (or base)
- Magnetic stirrer
- Buffer solutions for calibration
- Unknown solutions of equal concentration (e.g. $0.100 \text{ mol dm}^{-3}$ strong acid, weak acid, strong base, weak base, neutral salt)

Method (A — Identifying Unknown Solutions)

1. Calibrate the pH meter with standard buffers; rinse the probe with distilled water between samples.
2. Measure the pH of each unknown (all the same concentration). At $0.100 \text{ mol dm}^{-3}$: strong acid pH 1.0, weak acid roughly 2–4, neutral salt about 7, weak base roughly 10–12, strong base 13.0.
3. For a weak acid, use $[\text{H}^+] = 10^{-\text{pH}}$ and $K_a = [\text{H}^+]^2/[\text{HA}]$ to estimate K_a .

Method (B — pH Curve and Buffers)

1. Calibrate pH meter with standard buffers.
2. Pipette 25.0 cm^3 of acid into beaker. Record initial pH.
3. Add base from burette in small increments. Record pH after each addition.
4. Use $0.1\text{--}0.2\text{ cm}^3$ increments near the steep equivalence region.
5. Continue well past equivalence until pH plateaus.
6. Plot pH vs. volume to produce the titration curve.
7. **Buffer investigation:** Prepare buffer ($\text{HA} + \text{NaA}$). Add small amounts of HCl and NaOH . Show pH barely changes compared to water.

Safety

- NaOH — corrosive, goggles and gloves.
- pH probes — fragile, handle carefully.

Analysis

Identify: Equivalence point, half-equivalence point ($\text{pH} = \text{pK}_a$ for WA/SB), buffer region.

Select indicator: Must have colour change within the steep region of the curve.

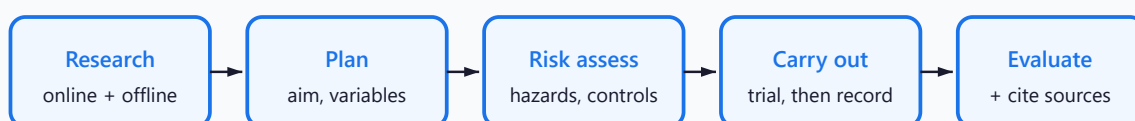
Error Analysis

Source of Error	Type	Improvement
Uncalibrated pH meter	Systematic	Calibrate with 2-3 buffers
Too-large increments near equivalence	Systematic	Use 0.1 cm^3 additions
pH not stabilised	Random	Wait for stable reading; stir continuously
CO_2 absorption	Systematic	Fresh NaOH ; keep stoppered

Exam Tip: For weak acid + strong base: initial $\text{pH} > 1$ (partially dissociated), buffer region before equivalence, equivalence $\text{pH} > 7$, half-equivalence $\text{pH} = \text{pK}_a$. The buffer region exists

because both HA and A^- are present in significant quantities. Use phenolphthalein for this combination.

PAG12: Research Skills



Example question (OCR's suggestion)

How long does it take an iron tablet to break down in the stomach?
Model the stomach with dilute HCl at $37\text{ }^{\circ}\text{C}$; measure Fe^{2+} released over time

PAG12 is not a single fixed experiment. It is an open-ended investigation in which you research a question yourself, plan and carry out your own method, and correctly reference the sources you used. It covers the CPAC competency on using online and offline research skills and citing sources.

Apparatus

- Depends on your plan. For the iron tablet example: iron(II) sulfate tablets, dilute HCl (to model stomach acid, pH about 1–2), a water bath at $37\text{ }^{\circ}\text{C}$, stirrer, stopwatch, pipettes
- For analysis of Fe^{2+} : burette and standard $\text{KMnO}_4(\text{aq})$ for redox titration, or a colorimeter with a suitable colour-forming reagent
- Access to textbooks, journals, reputable websites and data books for research

Method (Investigative Approach)

1. **Research:** find background information and possible methods from several sources (e.g. textbooks, the CLEAPSS/RSC resources your school uses, scientific articles). Note full details of each source as you go.

2. **Plan:** write an aim and a testable question. Identify the independent, dependent and control variables and choose apparatus with suitable resolution.
3. **Risk assessment:** list the hazards of each chemical and procedure and how you will reduce the risk.
4. **Trial and refine:** carry out a preliminary run to choose sensible quantities and sampling times.
5. **Carry out:** e.g. drop a tablet into the acid at 37 °C, withdraw samples at timed intervals, and titrate each with $\text{KMnO}_4(\text{aq})$ to find how much Fe^{2+} has dissolved (for a KMnO_4 titration, model the stomach acid with dilute H_2SO_4 rather than HCl , because MnO_4^- also oxidises Cl^- ions; or use colorimetry instead). Repeat to check reproducibility.
6. **Evaluate and reference:** compare your findings with published information and cite every source you used in a consistent referencing style.

Safety

- Your own written risk assessment must be checked by your teacher before you start.
- Dilute HCl and KMnO_4 are irritants — goggles; KMnO_4 is an oxidiser and stains.
- Iron tablets must not be taken internally or removed from the lab.

Analysis

Process data: plot the amount of Fe^{2+} released against time and find when release is complete (or 50% complete). Comment on the uncertainty in each measurement.

Referencing: for each source give the author(s), title, publisher or website, year and (for web sources) the URL and date accessed. Judge reliability: is it peer-reviewed, recent, and from a reputable organisation?

Error Analysis

Source of Error	Type	Improvement
Model does not match real conditions (no stirring/enzymes)	Systematic	Stir at a constant rate; state the limitations of the model
Fe^{2+} oxidised to Fe^{3+} by air before titration	Systematic	Titrate samples immediately; keep them acidified

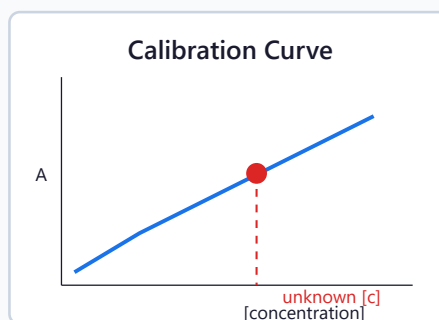
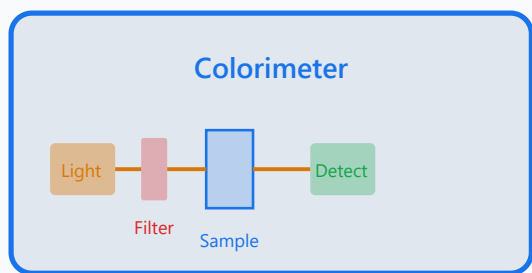
Tablets vary in mass/coating	Random	Use several tablets from the same batch and average
Unreliable or uncited sources	—	Use reputable sources and reference all of them

Exam Tip: Written papers can test PAG12 skills through planning questions: expect to write a method with named apparatus and quantities, identify variables to control, suggest a risk and its control measure, and evaluate whether a secondary source is reliable. Always justify your choices rather than just listing them.

Supporting Practicals (not PAGs)

The three practicals below are **not** Practical Activity Groups in the OCR Chemistry B (Salters) specification. They are kept because they practise examinable techniques: colorimetry can be used for continuous monitoring (PAG9), chromatography is one of the PAG6 techniques, and the equilibrium work supports the equilibrium content of the course.

Supporting: Colorimetry and Transition Metal Complexes



$$A = \epsilon cl \text{ (Beer-Lambert Law)}$$

Select complementary colour filter for the solution

Blue solution → use orange/red filter | Green solution → use red filter

Apparatus

- Colorimeter with selectable filters, cuvettes
- Standard solutions of known concentration (e.g., CuSO_4)
- Volumetric flasks for dilution
- Unknown concentration solution

Method

1. Select the appropriate colour filter (complementary to the solution colour).
2. Zero the colorimeter using distilled water as a blank.
3. Prepare at least 5 standard solutions of known concentration.
4. Measure the absorbance of each standard and the unknown.
5. Plot a calibration curve: absorbance (y) vs. concentration (x).
6. Read the unknown concentration from the graph using its absorbance value.

Safety

- Transition metal salts — wear gloves.
- Handle cuvettes by the top edges only.

Analysis

Plot: Calibration curve (should be linear at low concentrations per Beer-Lambert Law).

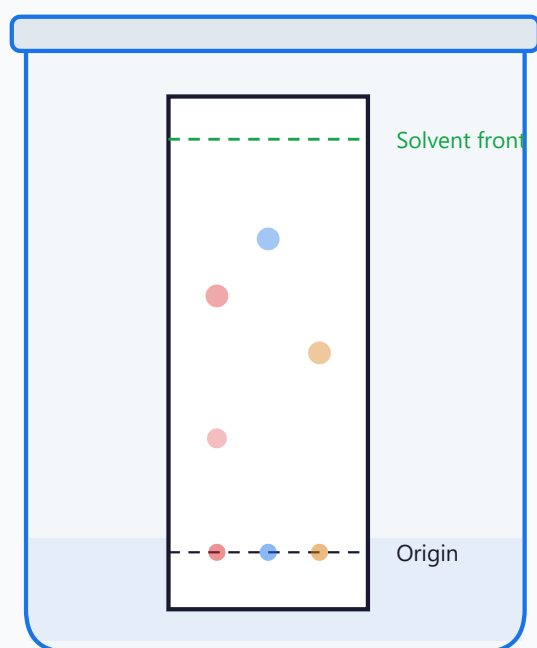
Read: Unknown concentration by interpolation (never extrapolation).

Error Analysis

Source of Error	Type	Improvement
Fingerprints/scratches on cuvette	Systematic	Handle by top; clean with lens tissue; same orientation
Wrong filter	Systematic	Use complementary colour; determine λ_{max} experimentally
Bubbles in solution	Random	Degas solutions; tap to remove bubbles
Standards incorrectly prepared	Systematic	Use volumetric flasks and pipettes

Exam Tip: The calibration curve must pass through the origin (zero concentration = zero absorbance after blanking). Only use values within the range of the calibration standards (interpolation, not extrapolation). If the curve is not linear at high concentrations, the Beer-Lambert Law breaks down — use only the linear portion.

Supporting: Chromatography



R_f Value

$$\frac{\text{distance moved by spot}}{\text{distance by solvent front}}$$

Paper chrom.

Partition between
mobile (solvent) and
stationary (water in
paper fibres) phase

Use pencil for origin line (ink dissolves). Spots must be above solvent level.

Apparatus

- TLC plates (silica or alumina) or chromatography paper
- Developing chamber (beaker with watch glass/lid)
- Capillary tubes, pencil, ruler
- Suitable solvent(s)
- UV lamp or ninhydrin/iodine for visualisation

Method

1. Draw a pencil origin line 1 cm from the bottom of the plate/paper.
2. Spot samples (and known references) onto the origin using capillary tubes.
3. Pour solvent into the chamber to 0.5 cm depth (below origin). Allow to saturate.
4. Place the plate/paper in the chamber. Cover. Let solvent rise.
5. Remove when solvent front is near the top. Mark the solvent front immediately.
6. Visualise spots (UV, iodine, or ninhydrin for amino acids).
7. Measure distances and calculate R_f values.

Safety

- Solvents are flammable — no flames, fume cupboard.
- UV lamp — avoid direct exposure to eyes.
- Ninhydrin — irritant, wear gloves.

Analysis

Calculate: $R_f = \text{distance by spot} / \text{distance by solvent}$. Compare with reference standards.

TLC: Adsorption chromatography — polar substances have lower R_f (interact more with polar stationary phase).

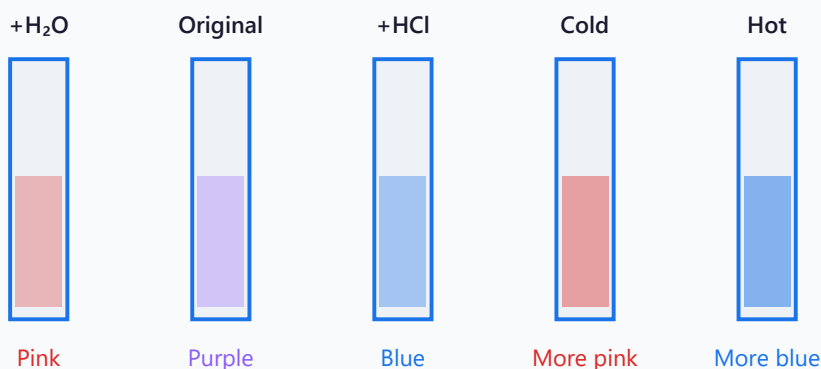
Paper: Partition chromatography — substances partition between water in paper and mobile solvent.

Error Analysis

Source of Error	Type	Improvement
Spots too large / overloaded	Systematic	Small concentrated spots; dry between applications
Origin below solvent level	Systematic	Check depth carefully
Chamber not saturated	Systematic	Line with filter paper; wait before inserting plate
Solvent front not marked in time	Random	Mark immediately on removal

Exam Tip: TLC uses adsorption (interaction with solid stationary phase). Paper chromatography uses partition (between two liquids). In TLC, more polar compounds have lower R_f because they adsorb more strongly to the polar silica. R_f values are only comparable under identical conditions.

Supporting: Equilibrium Investigation (Le Chatelier's)



Only temperature changes K_c . Concentration changes shift position, not K_c .

Apparatus

- Test tubes, test tube rack
- CoCl_2 solution (equilibrium mixture)
- Concentrated HCl, distilled water
- Water baths (ice and hot water)
- Teat pipettes

Method

1. Prepare cobalt chloride equilibrium mixture (purple at room temperature).
2. Divide into several test tubes.
3. **Add HCl:** Observe shift to blue (increased $[\text{Cl}^-]$ shifts equilibrium right).
4. **Add water:** Observe shift to pink (dilutes Cl^- , shifts equilibrium left).
5. **Cool in ice:** Observe shift to pink (exothermic direction favoured).
6. **Heat in hot water:** Observe shift to blue (endothermic direction favoured).

7. Record all observations and explain using Le Chatelier's principle.

Safety

- CoCl_2 is toxic and a suspected carcinogen — gloves.
- Conc. HCl is corrosive — fume cupboard.
- Hot water — care with burns.

Analysis

Record: Table of condition changed, direction of shift, colour observed, explanation.

Key: Forward reaction is endothermic. Heating shifts right (endothermic favoured). Adding Cl^- shifts right. Adding H_2O shifts left.

Error Analysis

Source of Error	Type	Improvement
Subjective colour judgement	Random	Use colorimeter for quantitative data
Temperature not maintained	Systematic	Monitor with thermometer; maintain bath
Equilibrium not re-established	Systematic	Allow sufficient time
Uncontrolled amounts added	Systematic	Measure volumes with pipette

Exam Tip: A catalyst does NOT shift the position of equilibrium and does NOT change K_c . It only increases the rate at which equilibrium is reached. In exam questions, always state that Le Chatelier's principle predicts the system opposes the change imposed. Only temperature changes K_c ; concentration and pressure changes only shift the position.